

Machine learning.

Introduction :- ML is the process of training a piece of Software, called model, to make useful predictions or generate content.

Types of ML systems

1. Supervised learning
2. Unsupervised learning
3. Reinforcement learning
4. Generative AI

1. Supervised learning :- An overview.

* Supervised learning models can make predictions after seeing lots of data with the correct answers and then discovering the connections between the elements in the data that produce the correct answers.

"Two of the most common use cases for supervised learning are Regression and Classification"

(i) Regression :- It predicts a numeric value. for example, a weather model that predicts the amount of rain, in inches or cm is a regression model.

(ii) Classification :- It predicts the likelihood that something belongs to a category. Unlike regression models, whose output is a number, classification models output a value that states whether or not something belongs to a particular category. for example it will predict whether an email is a spam or not.

Classification

Binary :- Have only two values
for eg:- Rain or No rain

Multiclass :- Contains more than two values

E.g:- rain, hail, snow or sleet.

2. Unsupervised learning :-

An unsupervised learning model aims to identify meaningful patterns in a dataset. For example, many unsupervised learning models rely on technique called Clustering to organize similar data into clusters (group).

"Clustering differs from classification because the categories aren't defined by you."

3. Reinforcement learning :-

It make predictions by getting rewards or penalties based on actions performed within an environment. A reinforcement learning system generates a policy that defines the best strategy for getting the most rewards.

4. Generative AI :-

Generative AI is a class of models that creates content from user input. For example generative AI can create unique images, music-compositions, jokes, it can summarize articles etc.

Gen-AI can take variety of inputs and create a variety of outputs.

Some Gen-AI models list:

Text-to-Text, Text-to-Image, Text-to-Video,
Text-to-Code, Text-to-Speech, Image and
Text-to-Image.